Test sheet: the dieresis U+0308 (and macron U+0304) in semibold italic

To-do item (2026-11-07): “Semibold italic dieresis U+0308 lands about 38 units left of the circumflex, breve and caron on every lowercase base tested (ȷ ı o n), and the macron U+0304 about 12 left: candidates for curated mark anchors, though the precomposed glyphs do not support moving the dieresis on lowercase.” All fonts are the patched fonts, except the two trial fonts in section 4.

1. Composed against precomposed, in numbers

For every lowercase precomposed letter of a Latin base with one above mark, the mark’s position relative to the base is compared: composed (base + CGJ + mark, shaped by HarfBuzz) minus precomposed (the designer’s glyph, found by matching the outlines of the base and of the mark). Δx is measured along the slant; positive means the composed mark lands right of the designer’s. Median over the letters (their number in brackets); ď ľ ť (comma caron) excluded. k is the mark’s own shift (the centre of its outline on its axis minus its anchor X): 0 in the anchor model, 150 in semibold italic (K). Where the precomposed mark is drawn differently (semibold italic macron and tilde), its ink centre is compared instead.

	Regular			Italic			Semibold			Semibold italic		
	k	Δx	Δy	k	Δx	Δy	k	Δx	Δy	k	Δx	Δy
◌̀ grave	-17	+20 (8)	-1	-16	+0.5 (8)	-1	-18	+9 (8)	-1	136	+17 (7)	+0
◌́ acute	32	+2 (17)	+0	32	+4 (17)	+0	36	-1 (17)	-1	187	+13.5 (16)	+7
◌̂ circumflex	-1	+3 (13)	-1	0	+18 (13)	-1	0	+0 (13)	-4	153	+16 (12)	-1
◌̃ tilde	0	-4 (8)	-4	11	+24 (8)	-4	-4	-9 (8)	+14	139	+12 (7)	-2
◌̄ macron	-2	-9 (7)	+18	-1	+0 (7)	-32	6	-2 (7)	-12	141	+5 (6)	-32
◌̆ breve	0	+8.5 (6)	-6.5	0	+5 (6)	-6.5	0	+6 (6)	-9.5	150	+14 (5)	-7
◌̈ dieresis	0	-0.5 (10)	-2	0	-3.5 (10)	+0	12	+13.5 (10)	-6	112	-26 (9)	+6
◌̊ ring	2	+3 (4)	+7.5	3	+8 (4)	+7	0	-0.5 (4)	+0.5	158	+24 (3)	+23
◌̌ caron	2	+5.5 (14)	+0	1	+10 (14)	+0	0	+0 (14)	+0	150	+1 (13)	+0

In semibold italic every mark lands 5–16 right of the precomposed glyphs except the dieresis, which lands 26 left: in the precomposed glyphs the dieresis sits where the other marks sit, about 40 right of where its anchor puts it. That is its k: 112 against 150. The macron’s k is 141, and its composed position is within 5 of the precomposed one (ink centres; the precomposed macron is drawn 32 higher). This contradicts the to-do item’s “the precomposed glyphs do not support moving the dieresis on lowercase”.

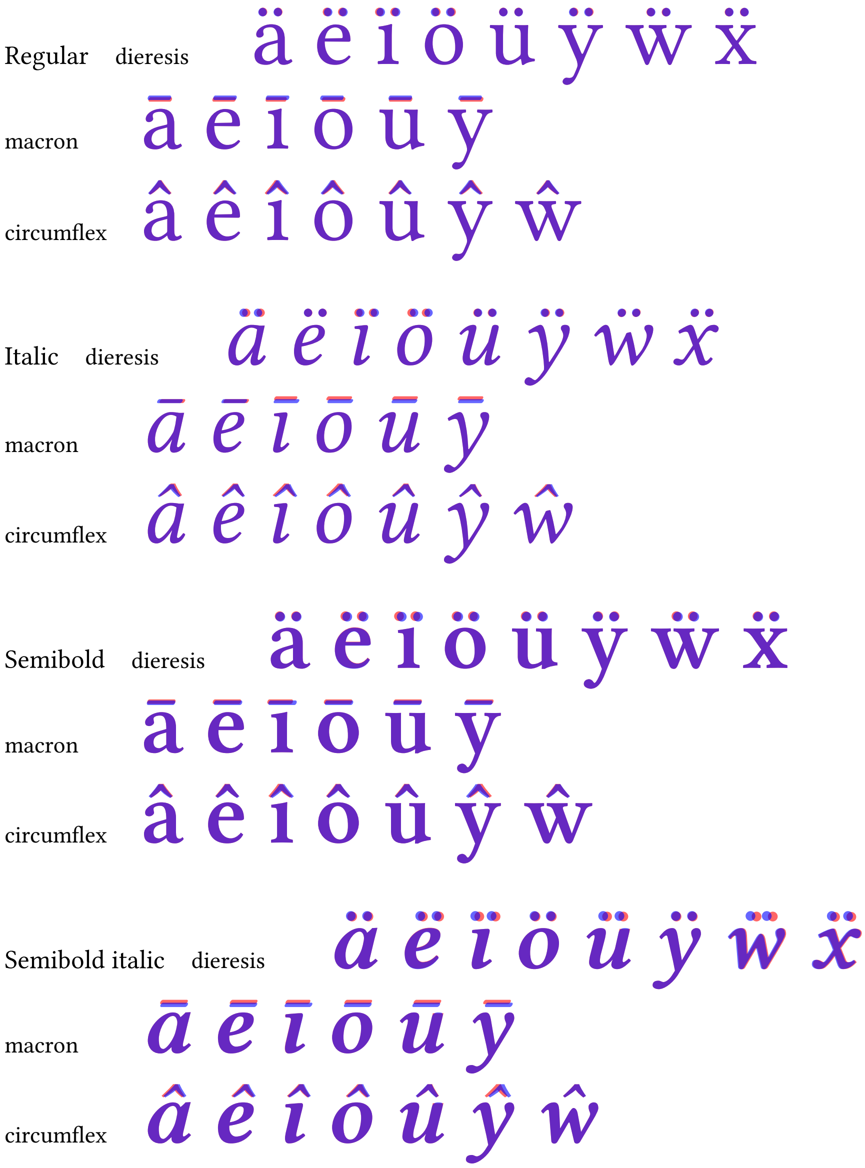
In the other styles: regular agrees with its precomposed glyphs throughout (within 10, except the grave); semibold’s dieresis ($k = 12$) lands 13.5 right of its precomposed glyphs, the other way; italic’s circumflex and tilde land 18 and 24 right, and its dieresis is 12–21 left of the circumflex relative to the precomposed glyphs on a, e, w.

Per base, the dieresis and the macron against the circumflex on the same base (each as composed minus precomposed); 0 means the two marks keep the relation they have in the precomposed glyphs:

	dieresis — circumflex							macron — circumflex						
	a	e	i	o	u	w	y	a	e	i	o	u	w	y
Regular	+1	+3	-2	+0	-7	-2	+4	-3	+0	-2	-4	-12	–	-4
Italic	-15	-12	-2	+4	+4	-21	-6	-14	-30	-5	-5	-3	–	+18
Semibold	+14	+16	+11	+12	+6	+11	+17	+5	+8	+7	+4	-7	–	+13
Semibold italic	-37	-49	-56	-27	–	-58	-50	-15	-14	+0	-15	–	–	-14

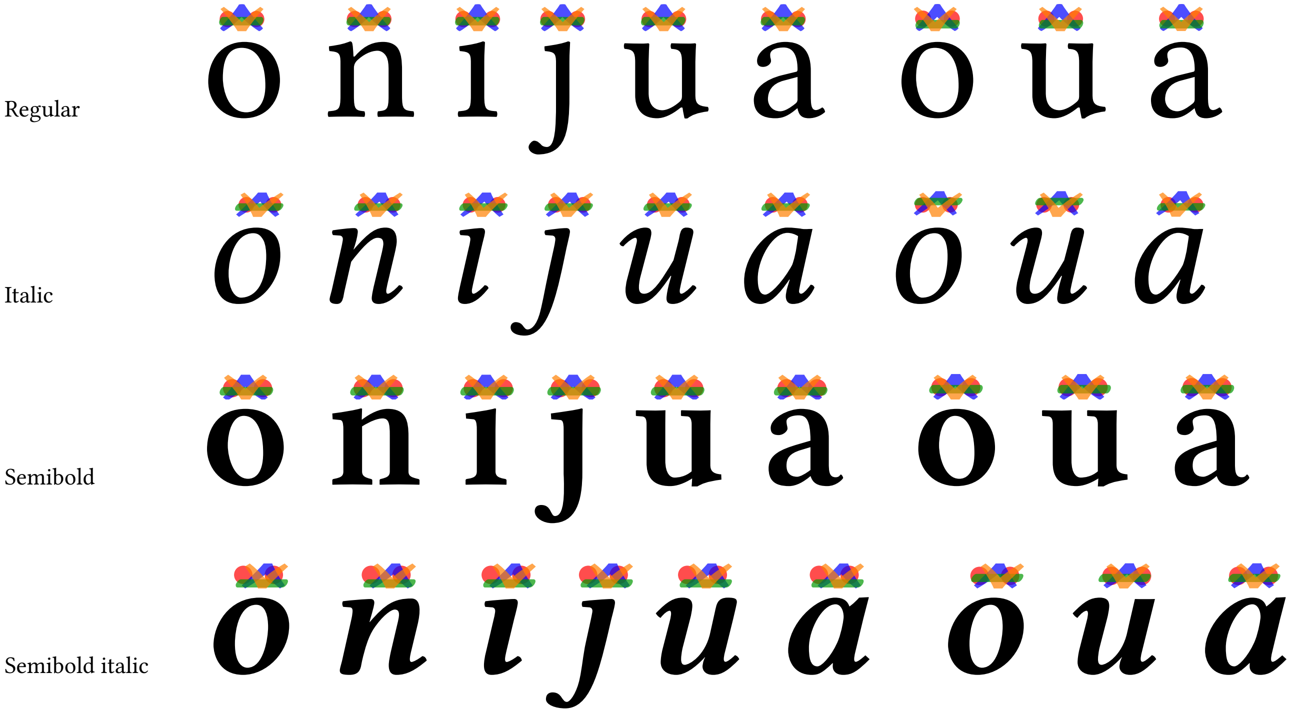
2. Composed (blue) over precomposed (red)

Purple where they coincide. The base is the same glyph in both, except where the designer moved it in the precomposed glyph (semibold italic: about 12 right).



3. The marks against each other on one base

Each cluster overlays four composed marks on one base: dieresis (red), circumflex (blue), macron (green), caron (orange). The last three clusters are the designer's precomposed ö ô ō ǒ, ü û ū ǔ, ä â ā ǎ, overlaid the same way.



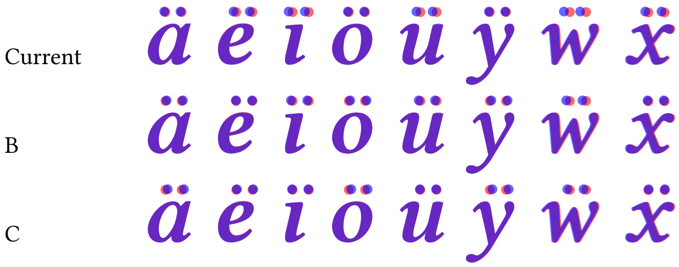
4. Options for semibold italic

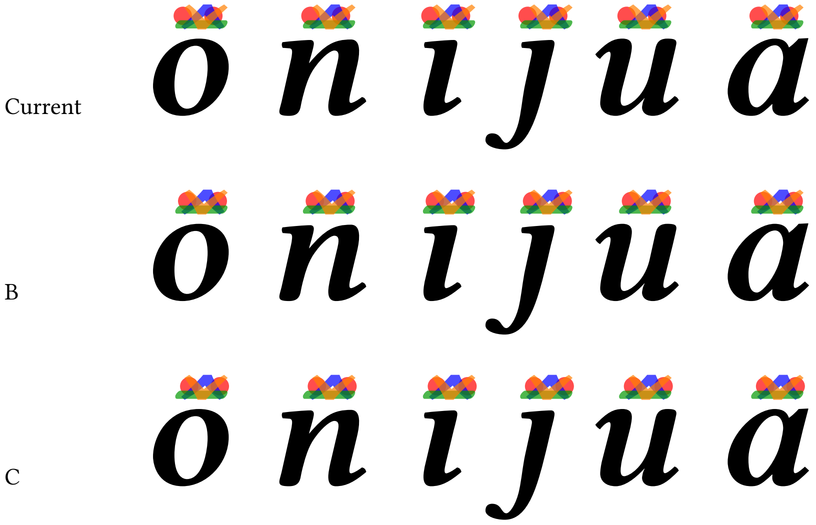
Current dieresis anchor $(-210, 645)$, regular's; $k = 112$.

B anchor $(-236, 645)$: the dieresis moves 26 right, onto the precomposed glyphs (median Δx 0).

C anchor $(-248, 645)$: the dieresis moves 38 right, $k = 150$ as the anchor model; it then keeps the other marks' relation to the precomposed glyphs (about 12 right of them, like the circumflex and breve).

Trial fonts B and C are the patched semibold italic with only that anchor changed.





5. In text

IPA centralized vowels mix precomposed letters (ä ë ö ü ï) with composed ones (ɛ̈ ɘ̈ ə̈ ɔ̈ ʊ̈ ɪ̈ ɐ̈ ʌ̈), so a dieresis off the precomposed position shows within one transcription. Same for the macron (ā ē ō ū against ɛ̄ ə̄ ɔ̄ ʊ̄).

Regular	[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]	[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]	naïve Zoë Brontë
Italic	*[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]*	*[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]*	*naïve Zoë Brontë*
Semibold	**[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]**	**[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]**	**naïve Zoë Brontë**
Semibold italic	***[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]***	***[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]***	***naïve Zoë Brontë***
SBI current	***[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]***	***[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]***	***naïve Zoë Brontë***
SBI B	***[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]***	***[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]***	***naïve Zoë Brontë***
SBI C	***[ä ɛ̈ ë ɘ̈ ə̈ ö ɔ̈ ü ʊ̈ ï ɪ̈ ɐ̈ ʌ̈]***	***[ā ɛ̄ ē ə̄ ō ɔ̄ ū ʊ̄]***	***naïve Zoë Brontë***